

Software Test Plan (STP)

Project: Hostel Management System

Version: 1.0

Team Number: 11

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*Reference Document: Hostel Management System – Software Requirements
Specification (SRS) v1.0, dated 05-09-2026*

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1. Introduction

1.1 Purpose

This document defines the Software Test Plan (STP) for the Hostel Management System (HMS) v1.0. It outlines the testing objectives, scope, test items, approach and strategy, entry and exit criteria, environment, resources, schedule, deliverables, risks, and responsibilities for testing the system.

This test plan is a direct continuation of the Hostel Management System Software Requirements Specification (SRS) v1.0. Every functional requirement (HMS-F-001 to HMS-F-017), non-functional requirement (HMS-NFR-001 to HMS-NFR-006), and security requirement (HMS-SEC-001 to HMS-SEC-006) defined in the SRS is carried forward into this plan and mapped to one or more test cases so that complete requirement coverage can be demonstrated.

1.2 Scope

Testing covers the Hostel Management System features described in the SRS, namely: administrator authentication, student record management (add, view, update, remove, search), hostel room management (add room, view room details, view available rooms, view occupied rooms), room availability maintenance, room allocation (allocate, prevent allocation of a full room, view allocated room, update allocation, vacate room), and the associated non-functional and security requirements.

Testing is performed at unit, integration, system, and acceptance levels on the web-based application built using Python (Django/Flask) with a MySQL database and an HTML/CSS/JavaScript frontend.

Excluded from testing are the internals of third-party components — the MySQL database engine, the Django/Flask framework, the web browser, and the operating system — as well as hardware maintenance and any functionality not specified in the SRS.

1.3 References

- Hostel Management System – Software Requirements Specification (SRS) v1.0, Team 11, dated 05-09-2026.
- Hostel Management System – UML Use-Case Diagrams (SRS Section 7).
- Hostel Management System – Requirements Traceability Matrix (SRS Section 8).
- Hostel Management System – Acceptance Tests (SRS Section 6.2, AT-001 to AT-016).
- IEEE 829 Standard for Software Test Documentation.
- Django / Flask framework documentation and MySQL documentation.

1.4 Definitions

Term	Definition
HMS	Hostel Management System
STP	Software Test Plan
SRS	Software Requirements Specification
RTM	Requirements Traceability Matrix
Admin	Authorized person responsible for managing hostel records and room allocation
Student	A student whose hostel and room details are maintained in the system
Room Allocation	The process of assigning an available hostel room to a student
Room Availability	Information about whether a hostel room has available capacity
FR	Functional Requirement
NFR	Non-Functional Requirement
TC	Test Case
AT	Acceptance Test
UAT	User Acceptance Testing
Defect	A deviation of the observed system behaviour from the behaviour specified in the SRS

2. Test Items

The following modules of the Hostel Management System v1.0, as identified in the SRS Requirements Traceability Matrix (SRS Section 8), are the items under test:

- Authentication module (administrator login, logout, session handling)
- Authorization / access control module (restriction of administrative operations)
- Student Management module (add, view, update, remove, search student records)
- Room Management module (add room details, view room details, view available and occupied rooms, maintain availability status)
- Room Allocation module (allocate room, prevent allocation of a full room, view allocated room, update allocation, vacate room)
- Data Validation module (validation of student and room input before storage)
- User Interface (login page, student details page, hostel and room management page, room availability page, room allocation page, student search page, room details page)

- Database layer (MySQL storage and retrieval of student, hostel, room, and allocation records)
- System performance and system architecture aspects covered by the non-functional requirements

3. Features to be Tested

The features below are mapped to the SRS requirement IDs. All high-priority requirements are mandatory for release; medium-priority requirements are tested in the same cycle but may be deferred only with the approval of the QA Lead and the Product Owner.

3.1 Functional Features

Req ID	Feature to be Tested	Priority
HMS-F-001	Administrator login using valid credentials	High
HMS-F-002	Rejection of login attempts with invalid credentials	High
HMS-F-003	Add new student details	High
HMS-F-004	View stored student details	High
HMS-F-005	Update student details	High
HMS-F-006	Remove a student record	Medium
HMS-F-007	Add hostel room details	High
HMS-F-008	View available hostel rooms	High
HMS-F-009	View occupied rooms	Medium
HMS-F-010	Allocate an available room to a student	High
HMS-F-011	Prevent allocation when the room has no available capacity	High
HMS-F-012	Display the room allocated to a student	High
HMS-F-013	Update a student's room allocation	Medium
HMS-F-014	Vacate a student's allocated room	Medium
HMS-F-015	Search for student records	Medium
HMS-F-016	Maintain current room availability status after allocation or vacation	High
HMS-F-017	View the details of a particular hostel room	Medium

3.2 Non-Functional Features

Req ID	Feature to be Tested	Type	Priority
HMS-NFR-001	Simple and user-friendly interface for common operations	Usability	High
HMS-NFR-002	Response to normal user requests within 3 seconds	Performance	High
HMS-NFR-003	Accurate and consistent student, room, and allocation information	Reliability	High
HMS-NFR-004	Availability during normal hostel administrative working hours	Availability	Medium
HMS-NFR-005	Maintainability of modules without affecting unrelated functions	Maintainability	Medium
HMS-NFR-006	Support for multiple student records without significant degradation	Scalability	Medium

3.3 Security Features

Req ID	Feature to be Tested	Priority
HMS-SEC-001	Valid login credentials required before accessing administrative functions	High
HMS-SEC-002	Rejection of invalid username or password combinations	High
HMS-SEC-003	Restriction of student and room management operations to authorized administrators	High
HMS-SEC-004	Protection of stored student and hostel information from unauthorized modification or deletion	High
HMS-SEC-005	Termination or restriction of access after logout	Medium
HMS-SEC-006	Validation of user input before storing student and room information	Medium

4. Features Not to be Tested

- Internal logic of the MySQL database engine (assumed tested by the vendor).
- Internal logic of the Django / Flask web framework and Python standard libraries (assumed tested by the vendor).
- Web browser internals and rendering engines of Google Chrome, Microsoft Edge, and Mozilla Firefox.

- Operating system functionality of Windows and Linux, and network infrastructure configuration.
- Hardware devices such as the computer, keyboard, mouse, monitor, and network hardware (SRS Section 3.2 requires no special hardware).
- Features outside the scope of the SRS, such as hostel fee management, mess management, attendance management, complaint or maintenance requests, leave or gate-pass management, e-mail/SMS notifications, online payment gateways, and mobile applications.
- Multi-hostel deployment across geographically distributed servers, load balancing, and disaster recovery, as these are not specified in the SRS.
- Data migration from any existing manual or legacy record system.

5. Test Approach / Strategy

5.1 Test Levels

- Unit tests (module-level): individual functions and views such as credential validation, student record creation, room capacity calculation, and availability update logic are tested in isolation using Python unittest / PyTest.
- Integration tests (Application ↔ Database, and module ↔ module): verification that the Student Management, Room Management, and Room Allocation modules interact correctly with the MySQL database and with each other — for example, that allocating a room correctly updates both the allocation record and the room availability status.
- System tests (end-to-end HMS functionality): complete workflows exercised through the browser, such as login → add student → add room → check availability → allocate room → view allocated room → vacate room → logout.
- Acceptance tests (UAT): execution of acceptance tests AT-001 to AT-016 defined in SRS Section 6.2 by the hostel administrator representative and the reviewing faculty.

5.2 Test Types

- Functional testing (core features HMS-F-001 to HMS-F-017), using positive, negative, and boundary test data.
- Regression testing after every defect fix and every new build, focused on the authentication, allocation, and availability paths.
- Performance testing (response time for common operations, verifying the 3-second limit of HMS-NFR-002, and behaviour as the number of student records grows for HMS-NFR-006).
- Usability testing (clarity of labels, input fields, buttons, and error messages, as required by HMS-NFR-001 and SRS Section 3.1).

- Reliability and data-consistency testing (HMS-NFR-003), verifying that records remain consistent after add, update, allocation, and vacation operations.
- Availability testing (HMS-NFR-004) during the defined hostel administrative working hours.
- Maintainability testing (HMS-NFR-005), verifying that a module can be modified without causing failures in unrelated modules.
- Compatibility testing across Google Chrome, Microsoft Edge, and Mozilla Firefox on Windows and Linux.
- Security testing (HMS-SEC-001 to HMS-SEC-006), detailed in Section 5.4.
- Database testing, verifying correct storage and retrieval of student, hostel, room, and allocation records.

5.3 Entry and Exit Criteria

Entry Criteria

- The SRS v1.0 has been reviewed and baselined.
- A stable build of the Hostel Management System has been delivered and deployed to the test environment.
- The test environment (application server, MySQL database, browsers, tools) is ready and accessible.
- Test cases and test data have been prepared and reviewed by the QA Lead.
- A smoke test of login and basic navigation passes on the delivered build.

Exit Criteria

- 100% of the planned test cases have been executed.
- All high-priority functional requirements (HMS-F-001, F-002, F-003, F-004, F-005, F-007, F-008, F-010, F-011, F-012, F-016) have been implemented and verified as Passed.
- All acceptance tests AT-001 to AT-016 defined in SRS Section 6.2 have been completed successfully.
- Zero critical and zero high-severity defects remain open; no critical security issue remains unresolved.
- The specified non-functional requirements are satisfied under normal operating conditions.
- Student, room, and room allocation information is stored and retrieved correctly.
- The Requirements Traceability Matrix confirms that every requirement has at least one executed test case and that the required test cases have been completed successfully.
- The Test Summary Report has been prepared and approved.

These exit criteria are aligned with the Exit Criteria stated in SRS Section 6.3.

5.4 Security Validation

- Validate credential handling: passwords masked on entry, not displayed, and not written to application logs (HMS-SEC-001).
- Verify rejection of invalid username and password combinations, including empty fields and wrong case (HMS-SEC-002).
- Verify authorization: attempt to access student and room management operations with a non-administrator or unauthenticated session and confirm access is denied (HMS-SEC-003).
- Verify data protection: attempt unauthorized modification or deletion of student, room, and allocation records through direct URL manipulation and confirm the attempt is prevented (HMS-SEC-004).
- Verify session termination after logout, including use of the browser Back button and direct entry of protected page URLs (HMS-SEC-005).
- Input validation and negative testing on all forms: blank mandatory fields, over-length values, invalid data types for room capacity and student identifiers, and special characters (HMS-SEC-006).
- Basic injection testing on login and search fields (SQL injection patterns) and basic cross-site scripting checks on student name and room description fields.
- Verification that the system prevents allocation of a room without available capacity even when the request is submitted directly, protecting record integrity (HMS-F-011, HMS-SEC-004).
- Verification that HTTP/HTTPS communication between the browser and the application server behaves as described in SRS Section 3.4, and that HTTPS is used when the system is deployed over a network.

6. Test Environment

6.1 Hardware

As stated in SRS Section 3.2, no special hardware is required. The test setup uses:

- Desktop or laptop computer with a minimum of 4 GB RAM and an Intel i3 (or equivalent) processor.
- Keyboard, mouse, and monitor/display.
- Local network or Internet connection, depending on the deployment mode.
- A separate machine acting as the client, used to verify multi-user and network access.

6.2 Software

Component	Details
Operating System	Windows 10/11 or Linux (Ubuntu)
Application	Hostel Management System v1.0
Backend	Python 3.x with Django or Flask framework
Frontend	HTML, CSS, JavaScript
Database	MySQL (test instance with dedicated test schema)
Web Browsers	Google Chrome, Microsoft Edge, Mozilla Firefox (latest stable versions)
Protocol	HTTP / HTTPS between browser and application server

6.3 Tools

Tool	Purpose
Selenium WebDriver	UI automation and regression testing of web pages
PyTest / Python unittest	Unit and integration testing of backend modules
Postman	Testing application endpoints and request/response validation
Apache JMeter	Performance testing — response time (HMS-NFR-002) and scalability (HMS-NFR-006)
MySQL Workbench	Verification of stored student, room, and allocation records
Browser Developer Tools	Inspection of requests, session behaviour, and UI defects
Jira / GitHub Issues	Defect logging, tracking, and status reporting
Git / GitHub	Version control for test scripts and test data
Microsoft Excel / Google Sheets	Maintaining the Requirements Traceability Matrix and test execution logs

6.4 Test Data

- Valid and invalid administrator login credentials.
- Sample student records containing all mandatory fields, plus records with missing, over-length, and invalid field values for negative testing.
- Sample hostel and room records with different capacities (for example, 1-seater, 2-seater, and 3-seater rooms).

- Rooms in three states: fully vacant, partially occupied, and fully occupied (to exercise HMS-F-008, HMS-F-009, and HMS-F-011).
- Existing allocation records required for update (HMS-F-013) and vacate (HMS-F-014) operations.
- A bulk data set of student records used to verify scalability (HMS-NFR-006).
- Search inputs covering exact matches, partial matches, and non-existent student records (HMS-F-015).

All test data is dummy data created for testing purposes only. No real student personal information is used in the test environment.

7. Test Schedule

Milestones:

Milestone	Planned Date / Duration	Owner
Test plan preparation and review	16-Sep-2026 to 17-Sep-2026	Pavithra G C
Test case design (functional, NFR, security)	18-Sep-2026 to 20-Sep-2026	Pavan Kumar H S, Prabhakar Kumar
Test environment setup and smoke test	21-Sep-2026	Pradeep
Unit testing	22-Sep-2026 to 25-Sep-2026	All members
Integration testing	26-Sep-2026 to 29-Sep-2026	Pavan Kumar H S
System testing (end-to-end)	30-Sep-2026 to 05-Oct-2026	All members
Security and performance testing	06-Oct-2026 to 08-Oct-2026	Prabhakar Kumar
Defect fixing and regression testing	09-Oct-2026 to 10-Oct-2026	All members
Acceptance testing (UAT, AT-001 to AT-016)	12-Oct-2026 to 14-Oct-2026	Pradeep, Pavithra G C
RTM update and Test Summary Report	15-Oct-2026	Pradeep
Final review and sign-off	16-Oct-2026	QA Lead and Product Owner

8. Test Deliverables

- Software Test Plan (this document)
- Test Cases — functional (TC-001 to TC-017), non-functional (TC-NFR-001 to TC-NFR-006), and security (TC-SEC-001 to TC-SEC-006)
- Acceptance Test Cases (AT-001 to AT-016, as defined in SRS Section 6.2)

- Test Scripts (Selenium and PyTest automation scripts)
- Test Data sets (student, hostel, room, and allocation records)
- Test Execution Logs
- Defect Reports
- Updated Requirements Traceability Matrix (with Status updated from N to P or A)
- Test Summary Report

9. Roles and Responsibilities

Responsibilities are assigned in line with the Division of Work agreed in the SRS document.

Role	Name	Responsibility
QA Lead	Pavithra G C (PES2UG24CS346)	Prepare and maintain the test plan, coordinate test execution, review test cases and results, consolidate the Test Summary Report
Test Engineer – Functional	Pavan Kumar H S (PES2UG24CS344)	Design and execute functional test cases for student management, room management, and room allocation (HMS-F-001 to HMS-F-017); log defects
Test Engineer – NFR & Security	Prabhakar Kumar (PES2UG24CS355)	Design and execute non-functional and security test cases (HMS-NFR-001 to HMS-NFR-006, HMS-SEC-001 to HMS-SEC-006); performance and usability measurement
Test Engineer – Traceability & UAT	Pradeep (PES2UG24CS357)	Set up the test environment, maintain the Requirements Traceability Matrix, execute acceptance tests AT-001 to AT-016, prepare execution logs
Developer	All four team members	Support defect triage, fix reported defects, deliver stable builds for regression testing
Product Owner / Reviewer	Course Faculty (Reviewer)	Approve test results and sign off on release readiness of the mini-project

10. Risks and Mitigation

Risk	Mitigation
Delay in delivery of a stable build	Request early partial/smoke builds; test completed modules first and integrate incrementally
Test environment or database server downtime	Maintain a local backup MySQL instance and a database dump so testing can continue offline
Incomplete or unavailable test data	Prepare dummy student, hostel, and room data in advance and store it as reusable SQL scripts

Risk	Mitigation
Requirement ambiguity during test design	Clarify against the SRS with the team and the faculty reviewer; record the clarification in the test case
Limited time due to parallel academic work and mini-project deadlines	Prioritise high-priority requirements first; execute medium-priority test cases in the second cycle
Room availability logic defects affecting multiple test cases (HMS-F-010, F-011, F-016)	Treat these as blocking defects, fix on priority, and run a focused regression suite after the fix
Loss of test artifacts or version confusion	Maintain all test cases, scripts, and reports in a shared GitHub repository with version control
Limited testing experience within the team	Use the acceptance tests from SRS Section 6.2 as a baseline and review test cases jointly before execution
Browser compatibility issues discovered late	Perform compatibility checks on Chrome, Edge, and Firefox from the start of system testing

11. Assumptions & Dependencies

- The Hostel Management System will be developed using the technologies stated in SRS Section 2.4 (Python with Django/Flask, MySQL, HTML/CSS/JavaScript).
- The MySQL test database will be available and stable throughout the test execution period.
- Dummy test data (student, hostel, room, and allocation records) will be provided before execution begins.
- Administrator test credentials will be created by the development team before the start of authentication testing.
- The requirements baselined in SRS v1.0 will not change significantly during the test execution period; any change will trigger an update to this test plan and to the RTM.
- All four team members will be available for test execution as per the schedule in Section 7.
- Testing depends on the availability of the application server and the database, as stated in SRS Section 2.5 (Constraints).
- The system is tested under normal operating conditions on a local network or single-machine deployment, consistent with the scope of the mini-project.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment (application server or MySQL database) is unavailable for more than 4 hours.

- The delivered build is too unstable and blocks more than 30% of the planned test cases.
- The administrator login functionality fails (HMS-F-001 / HMS-SEC-001), since all other administrative test cases depend on it.
- The room allocation or availability logic fails in a way that corrupts stored records (HMS-F-010, HMS-F-016).
- A critical security defect is found that exposes stored student data.
- Test data is lost or the test database schema becomes inconsistent.

Resume testing if:

- The blocking defects have been resolved and verified through a smoke test.
- The test environment and the database have been restored and stabilised.
- A corrected build has been deployed and the affected modules pass re-verification.
- Test data has been restored from the backup SQL scripts.
- The QA Lead confirms that fewer than 30% of test cases remain blocked.

13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM) ensures that every requirement in the SRS is mapped to at least one test case, and that every test case traces back to a requirement. The RTM defined in SRS Section 8 is carried forward here and extended with acceptance test references. The Status column uses the same legend as the SRS: N – Not Tested, P – Passed, A – Action Required.

13.1 Example Mappings

- HMS-F-001 (Administrator login) → TC-001, TC-SEC-001, AT-001
- HMS-F-010 (Allocate room) → TC-010, AT-008
- HMS-F-011 (Prevent allocation of a full room) → TC-011, AT-009
- HMS-NFR-002 (Response time ≤ 3 s) → TC-NFR-002, AT-014
- HMS-SEC-005 (Protected access after logout) → TC-SEC-005, AT-016

13.2 Requirements Traceability Matrix

Req ID	Requirement (Short)	Module	Test Case(s)	Acceptance Test	Status
HMS-F-001	Administrator login	Authentication	TC-001	AT-001	N
HMS-F-002	Invalid login handling	Authentication	TC-002	AT-002	N
HMS-F-003	Add student details	Student Management	TC-003	AT-003	N

Req ID	Requirement (Short)	Module	Test Case(s)	Acceptance Test	Status
HMS-F-004	View student details	Student Management	TC-004	AT-004	N
HMS-F-005	Update student details	Student Management	TC-005	AT-005	N
HMS-F-006	Remove student record	Student Management	TC-006	—	N
HMS-F-007	Add room details	Room Management	TC-007	AT-006	N
HMS-F-008	View available rooms	Room Management	TC-008	AT-007	N
HMS-F-009	View occupied rooms	Room Management	TC-009	—	N
HMS-F-010	Allocate room	Room Allocation	TC-010	AT-008	N
HMS-F-011	Prevent allocation of full room	Room Allocation	TC-011	AT-009	N
HMS-F-012	View allocated room	Room Allocation	TC-012	AT-010	N
HMS-F-013	Update room allocation	Room Allocation	TC-013	AT-011	N
HMS-F-014	Vacate room	Room Allocation	TC-014	AT-012	N
HMS-F-015	Search student records	Student Management	TC-015	AT-004	N
HMS-F-016	Update room availability	Room Management	TC-016	AT-012	N
HMS-F-017	View room details	Room Management	TC-017	—	N
HMS-NFR-001	User-friendly interface	User Interface	TC-NFR-001	—	N
HMS-NFR-002	Response time	System Performance	TC-NFR-002	AT-014	N
HMS-NFR-003	Data accuracy and consistency	Database	TC-NFR-003	AT-015	N
HMS-NFR-004	System availability	System	TC-NFR-004	—	N
HMS-NFR-005	Maintainability	System Architecture	TC-NFR-005	—	N

Req ID	Requirement (Short)	Module	Test Case(s)	Acceptance Test	Status
HMS-NFR-006	Scalability	Database / System	TC-NFR-006	—	N
HMS-SEC-001	User authentication	Authentication	TC-SEC-001	AT-013	N
HMS-SEC-002	Invalid credentials rejection	Authentication	TC-SEC-002	AT-002	N
HMS-SEC-003	Admin access restriction	Authorization	TC-SEC-003	AT-013	N
HMS-SEC-004	Data protection	Database / Security	TC-SEC-004	—	N
HMS-SEC-005	Protected access after logout	Authentication	TC-SEC-005	AT-016	N
HMS-SEC-006	Input validation	Data Validation	TC-SEC-006	—	N

13.3 Test Case Management

- Every test case records: Test Case ID, linked Requirement ID, objective, preconditions, test data, test steps, expected result, actual result, status, and the tester's name.
- Test case IDs follow the SRS convention: TC-001 to TC-017 for functional requirements, TC-NFR-001 to TC-NFR-006 for non-functional requirements, TC-SEC-001 to TC-SEC-006 for security requirements, and AT-001 to AT-016 for acceptance tests.
- Test cases are stored in a shared repository and reviewed by the QA Lead before execution.
- Each defect logged is linked to the failing test case and to the corresponding requirement ID, so that requirement coverage and defect status can be reported together.
- The RTM Status column is updated after every execution cycle from N (Not Tested) to P (Passed) or A (Action Required).

14. Test Metrics & Reporting

Metrics collected:

- Percentage of test cases executed
- Percentage of test cases passed / failed
- Defect density (defects per module)
- Defect severity distribution (critical, high, medium, low)
- Defect aging (time from log to closure)

- Requirement coverage (% of SRS requirements with at least one executed test case)
- Number of reopened defects after regression testing
- Average response time measured for common operations (against the 3-second target of HMS-NFR-002)

Reports:

- Daily test execution status (test cases executed, passed, failed, blocked)
- Defect status report with severity and ownership
- Requirement coverage report generated from the RTM
- Final Test Summary Report, stating whether the exit criteria in Section 5.3 and SRS Section 6.3 have been met

15. Approvals

Role	Name	Signature / Date
QA Lead	Pavithra G C	16-09-2026
Test Engineer – Functional	Pavan Kumar H S	16-09-2026
Test Engineer – NFR & Security	Prabhakar Kumar	16-09-2026
Test Engineer – Traceability & UAT	Pradeep	16-09-2026
Product Owner / Faculty Reviewer	Sheela Mam	

Division of Work

The responsibilities for preparing this Software Test Plan were divided equally among the four team members, following the same distribution used for the SRS document.

Sl. No.	Team Member	Work Assigned
1	Pavithra G C	Introduction, Scope, References, Definitions, and integration and review of the complete test plan
2	Pavan Kumar H S	Test Items, Features to be Tested, Features Not to be Tested, and Test Environment
3	Prabhakar Kumar	Test Approach and Strategy, Security Validation, Risks and Mitigation, and Suspension & Resumption Criteria
4	Pradeep	Test Schedule, Test Deliverables, Traceability Matrix, Test Metrics & Reporting, and Approvals